

Artificial Intelligence Based Control

Lecture Notes

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Reference:

Dr. Lantos Béla: Fuzzy systems and genetic algorithms, 2001, Műegyetemi kiadó, Budapest

AI belong to soft computing.

Effective if only limited information is available about the system or the system is complicated, mathematical treatment is questionable.

Strong connection with the optimization and approximation. Intelligent approaches instead of exact description.

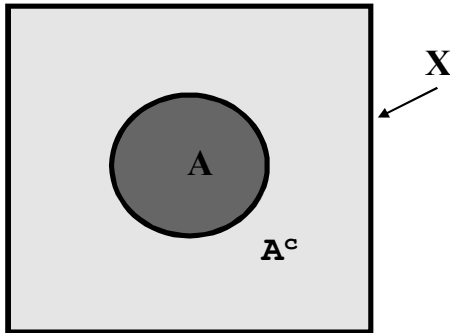
Application: Almost everywhere. Industry, robot control, Washing machines, Household appliances, vehicles (cars), HDD, etc.

1. Fundamentals of Fuzzy systems

1.1. Fuzzy sets and membership functions

1.1.1. Crisp sets

In classical set theory it is usual to talk about a subset A of a universe of discourse X , in which an arbitrary element $x \in X$ is either an element of A , or definitely not an element of A .



We can assign to set A its characteristic function $\kappa_A : X \rightarrow \{0,1\}$, which can take only 0 or 1 (crisp) numerical values:

$$\kappa_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{otherwise} \end{cases}$$

Set A can be identified with the pair (X, κ_A) : $A \leftrightarrow (X, \kappa_A)$

1.1.2. Fuzzy sets

1.1 Problem: The human reasoning often operates with uncertain linguistic expressions.

1.1. Example: a temperature θ that it is cold, cool, comfortable, warm or hot, and not everybody mean precisely the same.

Suggestion: a fuzzy set can be attached to the “cool” linguistic variable, which is defined by the function $\mu_{\text{cool}} : [-15^\circ\text{C}, -35^\circ\text{C}] \rightarrow [0,1]$, which is called the membership function of the fuzzy set and it says possibility of to be cool.

If X is the universe of discourse and A is a fuzzy set on X , then the degree to which an element $x \in X$ belongs to A is defined by $\mu_A(x) \in [0,1]$. The fuzzy set A can be identified with the pair: $A \leftrightarrow (X, \mu_A)$

definition of Fuzzy sets

Discrete universe of discourse: $A = \sum_{i=1}^n \mu_A(x_i) / x_i$ Symbolic sum

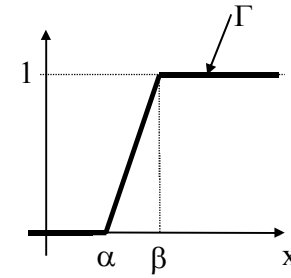
or a table

Continuous universe of discourse: $A = \int_{x \in X} \mu_A(x) / x$ Symbolic integral

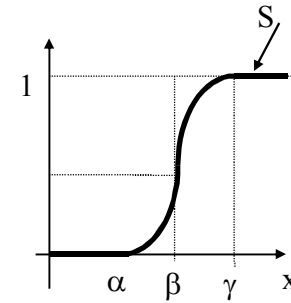
Symbolic notation in both cases.

1.1.3. Typical membership functions

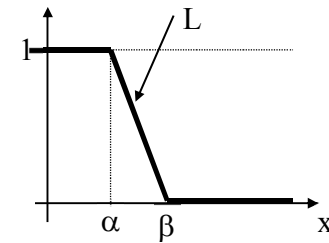
$$\Gamma(x, \alpha, \beta) = \begin{cases} 0, & \text{if } x < \alpha \\ \frac{x - \alpha}{\beta - \alpha}, & \text{if } x \in [\alpha, \beta] \\ 1, & \text{if } x > \beta \end{cases}$$



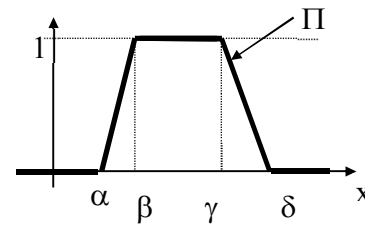
$$S(x, \alpha, \beta = \frac{\alpha + \gamma}{2}, \gamma) = \begin{cases} 0, & \text{if } x < \alpha \\ 2 \left(\frac{x - \alpha}{\gamma - \alpha} \right)^2, & \text{if } x \in [\alpha, \beta] \\ 1 - 2 \left(\frac{x - \gamma}{\gamma - \alpha} \right)^2, & \text{if } x \in [\beta, \gamma] \\ 1, & \text{if } x > \gamma \end{cases}$$



$$L(x, \alpha, \beta) = 1 - \Gamma(x, \alpha, \beta) = \begin{cases} 1, & \text{if } x < \alpha \\ \frac{\beta - x}{\beta - \alpha}, & \text{if } x \in [\alpha, \beta] \\ 0, & \text{if } x > \beta \end{cases}$$

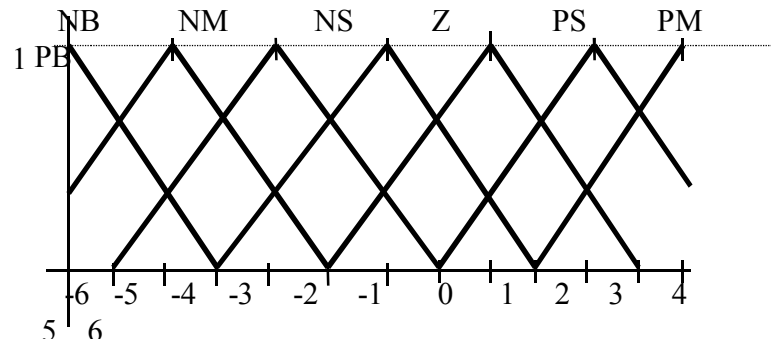


$$\Pi(x, \alpha, \beta, \gamma, \delta) = \begin{cases} 0, & \text{if } x < \alpha \\ \frac{x - \alpha}{\beta - \alpha}, & \text{if } x \in [\alpha, \beta] \\ 1, & \text{if } x \in [\beta, \gamma] \\ \frac{\delta - x}{\delta - \gamma}, & \text{if } x \in [\gamma, \delta] \\ 0, & \text{if } x > \delta \end{cases}$$



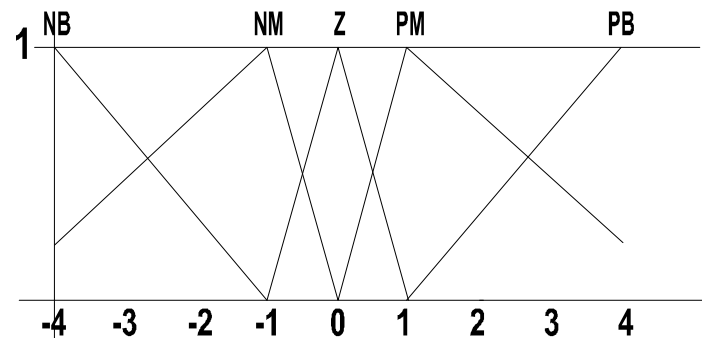
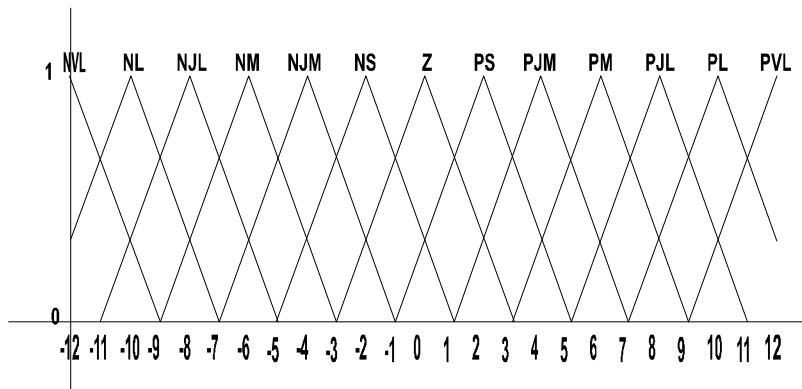
1.1.4. Normalization of physical variables

We often normalize the physical variable to a region and distinguish the fuzzy sets (linguistic values) by the sign (P=positive, N=negative) and magnitude (B=big, M=medium, S=small, Z=zero):



Standard region with 7 levels (equipartition)

Different resolutions and distributions:



1.2 Fuzzy operations and norms

1.2.1 Fuzzy set operations

Given two fuzzy sets, A and B . Based on membership functions $\mu_A(x)$ and $\mu_B(x)$ one can determine the membership function of the new fuzzy set.

- **Fuzzy union ($\cup$)**

Expected: If the possibility that x belongs to A or B are $\mu_A(x)$ and $\mu_B(x)$, then it is required that the possibility that is x belongs to the union is not decreased in case of $\mu_{A \cup B}(x)$ (S-norms).

$$\mu_{A \cup B}(x) = S(\mu_A(x), \mu_B(x)) = \mu_A(x) \vee \mu_B(x)$$

Example: $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$

- **Fuzzy intersection ($\cap$)**

- Expected: If the possibility that x belongs to A or B are $\mu_A(x)$ and $\mu_B(x)$, then it is required that the possibility that is x belongs to the union is not decreased in case of $\mu_{A \cap B}(x)$ (T-norms).

$$\mu_{A \cap B}(x) = T(\mu_A(x), \mu_B(x)) = \mu_A(x) \wedge \mu_B(x)$$

Example: $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$

- **Fuzzy complements**

Leggyakoribb: $\mu_{A^c}(x) = 1 - \mu_A(x)$

Különböző szabályok terjedtek el. (c-normák).

1.2.2 Fuzzy direct product

Let X_i denote the base set, let A_i denote a fuzzy set with membership function $\mu_{A_i}(x_i)$, where $A_i = (X_i, \mu_{A_i})$, $i = 1, 2, \dots, n$. Since $x = (x_1, \dots, x_n)$ contains $x_1, \dots, x_n$ simultaneously, the membership function of the fuzzy direct product $A_1 \times A_2 \times \dots \times A_n$ is:

$$\mu_{A_1 \times \dots \times A_n}(x_1, \dots, x_n) = \mu_{A_1}(x_1) \wedge \dots \wedge \mu_{A_n}(x_n)$$
$$A_1 \times \dots \times A_n \text{ fuzzy direct product} \leftrightarrow (X_1 \times \dots \times X_n, \mu_{A_1 \times \dots \times A_n})$$

Here $\wedge$ is appropriate T-norm, e.g: $\wedge = \min$.

1.2.3 Fuzzy relation

In the classical meaning, the binary relation $x R y$ (for instance $x \equiv y$ or $x \leq y$) specifies that some elements $(x, y) \in X \times X$ of the set are in relation R with each other, while the relation is not fulfilled by other $(x, y) \in X \times X$ elements.

(Classic) relation R : special $R \subset X \times X$ subset of the $X \times X$ direct product i.e

$$x R y \leftrightarrow (x, y) \in R \subset (X \times X)$$

Fuzzy relation:

$$\begin{aligned} \mu_R : X_1 \times \cdots \times X_n &\rightarrow [0, 1] \\ R \text{ fuzzy reláció} &\leftrightarrow (X_1 \times \cdots \times X_n, \mu_R) \end{aligned}$$

$\mu_R(x_1, \dots, x_n)$ gives the possibility that a relation between variables $x_1, \dots, x_n$ holds.

Question: Fuzzy relation defines a fuzzy set. Are there intersection and union between relations?

Answer: Yes, but only if they are defined on the same universe of discourse (UoD).

Question: What if the relations are defined on different UoD?

Answer: Cylindrical extension!

1.2.4. Cylindrical extension

Let R be a fuzzy relation with arguments on the product space $X_{i_1} \times X_{i_2} \times \cdots \times X_{i_r}$, which we want to extend to the product space $X_1 \times \cdots \times X_n$, where $\{i_1, \dots, i_r\} \subset \{1, \dots, n\}$:

$$\mu_{ce(R)}(x_1, \dots, x_n) := \mu_R(x_{i_1}, x_{i_2}, \dots, x_{i_r})$$

$$ce(R) = (X_1 \times \cdots \times X_n, \mu_{ce(R)}) = \int_{X_1 \times \cdots \times X_n} \mu_R(x_{i_1}, \dots, x_{i_r}) / (x_1, \dots, x_n).$$

1.2.5. Projection

Motivation. The aggregated relation of fuzzy control is defined on $X_1 \times \cdots \times X_n \times Y$ where $X_1 \times \cdots \times X_n$ is the input of the controller Y is the output of the controller. To know the effect on the output, one has to project the aggregated relation to set Y .

Let R be a fuzzy relation on the product space $X_1 \times \cdots \times X_n$, then its projection to the $X_{i_1} \times \cdots \times X_{i_r}$ space is a fuzzy relation $P = \text{Proj}_{X_{i_1} \times \cdots \times X_{i_r}}(R)$, whose membership function, with the notation $\{j_1, \dots, j_{n-r}\} = \{1, \dots, n\} \setminus \{i_1, \dots, i_r\}$, is:

$$\mu_P(x_{i_1}, \dots, x_{i_r}) := \sup_{x_{j_1}, \dots, x_{j_{n-r}}} \mu_R(x_1, \dots, x_n)$$

$$\text{Proj}_{X_{i_1} \times \cdots \times X_{i_r}}(R) = (X_{i_1} \times \cdots \times X_{i_r}, \mu_P)$$

In the case of finite base set or compact (bounded and closed) base set $\sup = \max$.

1.2.6. Fuzzy join

The fuzzy join is used for fitting contiguous or disjoint relations, and also plays a significant role in fuzzy logic operations. Let R and S denote fuzzy relations.

Let

R fuzzy relation on domain $X_1 \times X_2 \times \cdots \times X_r$,

S fuzzy relation on domain $X_m \times X_{m+1} \times \cdots \times X_n$, $m \leq r+1$.

The fuzzy join $\text{Join}(R, S)$ is on domain $X_1 \times X_2 \times \cdots \times X_n$.

$m < r+1$: the relations are intersecting

$m = r+1$: the relations are disjoint

We can obtain fuzzy join by cylindrical extension of both relations to the product space $X_1 \times \cdots \times X_n$ then considering the intersection of the extended relations:

$$\text{Join}(R, S) = \text{ce}(R) \cap \text{ce}(S)$$

$$\mu_{\text{Join}(R, S)}(x_1, \dots, x_n) := \mu_R(x_1, \dots, x_r) \wedge \mu_S(x_m, \dots, x_n)$$

$$\text{Join}(R, S) = \int_{X_1 \times \cdots \times X_n} \{ \mu_R(x_1, \dots, x_r) \wedge \mu_S(x_m, \dots, x_n) \} / (x_1, \dots, x_n).$$

Any T -norms are allowed, e.g: $\wedge = \min$

1.2.7. Fuzzy composition

At the algorithmization of fuzzy implications, the fuzzy composition plays an important role.

Let

R fuzzy relation on product space $X_1 \times \cdots \times X_{m-1} \times X_m \times \cdots \times X_r$,

S fuzzy relation on product space $X_m \times \cdots \times X_r \times X_{r+1} \times \cdots \times X_n$,

Then fuzzy composition $R \circ S$ is on product space $X_1 \times \cdots \times X_{m-1} \times X_{r+1} \times \cdots \times X_n$

We can obtain fuzzy composition by cylindrically extending both relations to the $X_1 \times \cdots \times X_n$ product space, then considering the intersection of the extended relations and project it to the product space $X_1 \times \cdots \times X_{m-1} \times X_{r+1} \times \cdots \times X_n$:

$$R \circ S = \text{Proj}_{X_1 \times \cdots \times X_{m-1} \times X_{r+1} \times \cdots \times X_n} [(\text{ce}(R) \cap \text{ce}(S))] = \text{Proj}[\text{Join}(R, S)]$$
$$\mu_{R \circ S}(x_1, \dots, x_{m-1}, x_{r+1}, \dots, x_n) := \sup_{x_m, \dots, x_r} \mu_R(x_1, \dots, x_r) \wedge \mu_S(x_m, \dots, x_n)$$

Any T -norms are allowed, e.g: $\wedge = \min$

Compostion can be considered matching of two fuzzy sets.

(It says how much the two relation holds at the same time at an element of product space)

First, we extend the two fuzzy sets (ce), then match them on their common product space (sup)

Analogy: Input data matching to a rule base

1.3 Fuzzy logic

By using the conjunction, disjunction and negation operations (commonly used in binary logic) we can logically relate linguistic variables, to which we can assign fuzzy sets to characterize their uncertainty.

1.3.1. *Fuzzy conjunction, disjunction and negation*

Let x be the linguistic variable, let X be the base set, and let A, B be fuzzy sets assigned to the linguistic variable on X with membership functions μ_A and μ_B respectively, then the results of fuzzy conjunction, disjunction and negation are fuzzy sets:

Conjunction: x is A and x is B
$$\mu_{A \cap B}(x) = \mu_A(x) \wedge \mu_B(x)$$

Disjunction: x is A or x is B
$$\mu_{A \cup B}(x) = \mu_A(x) \vee \mu_B(x)$$

Negation: x is not A
$$\mu_{A^c}(x) = c(\mu_A(x))$$

1.3.2. Fuzzy conjunction, disjunction and negation on different domains

Let $x_1, \dots, x_N$ be elementary variables, and let $X_1, \dots, X_N$ the corresponding base sets, furthermore let $\{i_1, \dots, i_r\}$ and $\{j_1, \dots, j_m\}$ be subsets of the index set $\{1, \dots, N\}$, let $X_A = X_{i_1} \times \dots \times X_{i_r}$ and $X_B = X_{j_1} \times \dots \times X_{j_m}$ be product sets, let $x_A \in X_A$, $x_B \in X_B$ be aggregated variables and $A \subset X_A$, $B \subset X_B$ be fuzzy sets on the product spaces. Then the general forms of conjunction and disjunction are the following.

Then

Conjunction: x_A is A and x_B is $B = v$ is P
 $P = \text{ce}(A) \cap \text{ce}(B)$
 $\mu_P(v) = \mu_A(x_A) \wedge \mu_B(x_B)$

Disjunction: x_A is A or x_B is $B = v$ is P
 $P = \text{ce}(A) \cup \text{ce}(B)$
 $\mu_P(v) = \mu_A(x_A) \vee \mu_B(x_B)$

1.3.3. Fuzzy implication

There are several methods for implications widely used in practice.

Implication in binary logic ($a \rightarrow b$):

a	b	$a \rightarrow b$
0	0	1
0	1	1
1	0	0
1	1	1

Zadeh implication:

$$a \rightarrow b = a \cdot b + \bar{a}$$

Kleene-Dienes implication:

$$a \rightarrow b = \overline{a \cdot \bar{b}} = \bar{a} + b$$

Mamdani implication:

$$a \rightarrow b \approx a \cdot b$$

Not compatible with Boole algebra!

1.3.4. Fuzzy knowledge base

Mamdani implication is the most popular!

Let $x \in X$, fuzzy set $A \subset X$ and $y \in Y$, $B \subset Y$ fuzzy set

Implication: *if x is A then y is B* (“else” is too complicated to use)

Mamdani implication:

$$\begin{aligned} A \rightarrow B &= \text{ce}(A) \cap \text{ce}(B) \\ \mu_{A \rightarrow B}(x, y) &= \mu_A(x) \wedge \mu_B(y) \\ A \rightarrow B &= \int_{X \times Y} \mu_A(x) \wedge \mu_B(y) / (x, y) \end{aligned}$$

i.e $R = A \rightarrow B \subset X \times Y$ is a relation on $X \times Y$.

Projection relation $R = A \rightarrow B$ to fuzzy set Y one can get the **composition rule of fuzzy implication**:

$$\text{Proj}_Y(R) = \text{Proj}_Y(A \rightarrow B) = A \circ B$$

where $A \circ B$ is the fuzzy composition defined earlier.

The general form of fuzzy implication with rule base (relations) $R_1, \dots, R_n$ (i.e the union $R = \bigcup_{i=1}^n R_i$) is given by

$$R_i: \quad \text{if } x_1 \text{ is } X_1^i \text{ and } \dots \text{ and } x_N \text{ is } X_N^i \text{ then } y \text{ is } Y^i$$

Using fuzzy control $R_i, i = 1, \dots, n$ is the rule base (knowledge base of fuzzy system) in the form of if...then implications

1.3.5. Data matching for a single relation

Let x_j^* is the (crisp) measured value of variable x_j and fuzzy set is A_j is the noisy measurement of variable x_j . In case of relation R_i , let fuzzy set A_j be defined on domain X_j^i .

In this case

The cylindrical extension of (measured) input data: $D = \text{ce}(A_1) \cap \dots \cap \text{ce}(A_N)$
 $\mu_D(x_1, \dots, x_N, y) = \mu_{A_1}(x_1) \wedge \dots \wedge \mu_{A_N}(x_N)$

The fuzzy form of Mamdani implication R_i (as a relation): $R_i = \text{ce}(X_1^i) \cap \dots \cap \text{ce}(X_N^i) \cap \text{ce}(Y^i)$

$$\mu_{R_i}(x_1, \dots, x_N, y) = \mu_{X_1^i}(x_1) \wedge \dots \wedge \mu_{X_N^i}(x_N) \wedge \mu_{Y^i}(y)$$

The composition rule of fuzzy implication (exploiting that $\text{ce}(A_j) \cap \text{ce}(B_j) = \text{ce}(A_j \cap B_j)$ on the same domain):

$$\begin{aligned} D \circ R_i &= \text{Proj}_Y(D \cap R_i) \\ \mu_{D \circ R_i}(y) &= \sup_{x_1, \dots, x_N} \mu_D(x_1, \dots, x_N, y) \wedge \mu_{R_i}(x_1, \dots, x_N, y) = \\ &= \sup_{x_1, \dots, x_N} [\mu_{A_1}(x_1) \wedge \dots \wedge \mu_{A_N}(x_N)] \wedge [\mu_{X_1^i}(x_1) \wedge \dots \wedge \mu_{X_N^i}(x_N) \wedge \mu_{Y^i}(y)] = \\ &= \sup_{x_1, \dots, x_N} [\mu_{A_1}(x_1) \wedge \mu_{X_1^i}(x_1)] \wedge \dots \wedge [\mu_{A_N}(x_N) \wedge \mu_{X_N^i}(x_N)] \wedge \mu_{Y^i}(y) = \\ &= [\sup_{x_1} \mu_{A_1}(x_1) \wedge \mu_{X_1^i}(x_1)] \wedge \dots \wedge [\sup_{x_N} \mu_{A_N}(x_N) \wedge \mu_{X_N^i}(x_N)] \wedge \mu_{Y^i}(y) \end{aligned}$$

Introducing

and the firing values $\tau_{ij} := \sup_{x_j} \mu_{A_j}(x_j) \wedge \mu_{X_j^i}(x_j)$

$$\tau_i := \tau_{i1} \wedge \cdots \wedge \tau_{iN},$$

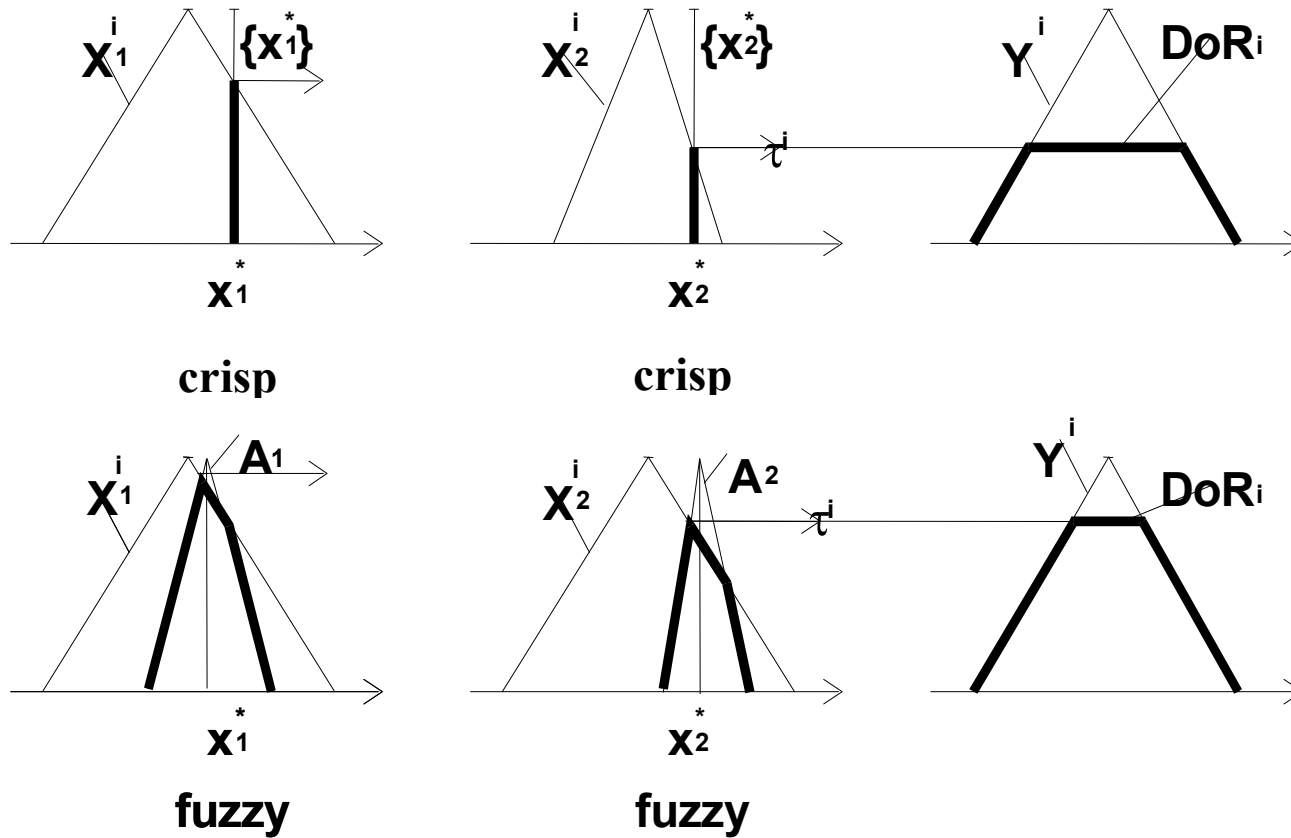
Then:

$$\mu_{D \circ R_i}(y) = \tau_i \wedge \mu_{Y^i}(y)$$

Most popular cases:

T=min: $\tau_{ij} := \sup_{x_j} \min\{\mu_{A_j}(x_j), \mu_{X_j^i}(x_j)\}, \quad \tau_i := \min\{\tau_{i1}, \dots, \tau_{iN}\},$

T=dot: $\tau_{ij} := \sup_{x_j} \mu_{A_j}(x_j) \cdot \mu_{X_j^i}(x_j), \quad \tau_i := \tau_{i1} \cdot \dots \cdot \tau_{iN}.$



1.3.6. Composition based inference (general case)

The rule base: $R = \bigcup_{i=1}^n R_i$

Relations: $R_i : \underbrace{\text{if } x_1 \text{ is } X_1^i \text{ and } \dots \text{ and } x_N \text{ is } X_N^i}_{\text{antecedent}} \text{ then } \underbrace{y \text{ is } Y^i}_{\text{consequent}}$

Antecedent:

$$B^i := \text{ce}(X_1^i) \cap \dots \cap \text{ce}(X_N^i)$$

$$\mu_{B^i}(x_1, \dots, x_N, y) = \mu_{X_1^i}(x_1) \wedge \dots \wedge \mu_{X_N^i}(x_N)$$

Implication:

$$R_i = B^i \rightarrow Y^i$$

$$\mu_{R_i}(x_1, \dots, x_N, y) = \mu_{B^i \rightarrow Y^i}(x_1, \dots, x_N, y)$$

Observation/Measurement (data):

$$x_j^* \rightarrow \text{crisp } A_j \text{ or fuzzy } A_j$$

$$D = \text{ce}(A_1) \cap \dots \cap \text{ce}(A_N)$$

$$\mu_D(x_1, \dots, x_N, y) = \mu_{A_1}(x_1) \wedge \dots \wedge \mu_{A_N}(x_N)$$

Composition based inference:

$$D \circ R = D \circ \bigcup_{i=1}^n R_i = \text{Proj}_Y \left\{ D \circ \bigcup_{i=1}^n R_i \right\}$$
$$\mu_{D \circ R}(y) = \sup_{x_1, \dots, x_N} \mu_D(x_1, \dots, x_N, y) \wedge \left\{ \bigvee_{i=1}^n \mu_{B^i \rightarrow Y^i}(x_1, \dots, x_N, y) \right\}$$

In general case, the computation of $\mu_{D \circ R}(y)$ is difficult since the order of $\sup, \wedge, \vee, B^i \rightarrow Y^i$ cannot be changed.
However...

Mamdani implication S=max, T=min or T=min - the algorithm is simple!

$$\begin{aligned}
 \mu_{D \circ R}(y) &= \sup_{x_1, \dots, x_N} \mu_D(x_1, \dots, x_N, y) \wedge \left\{ \bigvee_{i=1}^n [\mu_{B^i}(x_1, \dots, x_N, y) \wedge \mu_{Y^i}(y)] \right\} = \\
 &= \sup_{x_1, \dots, x_N} \min \{ \min \{ \mu_{A_1}(x_1), \dots \}, \max_{1 \leq i \leq n} \{ \min \{ \min \{ \mu_{X_1^i}(x_1), \dots \}, \mu_{Y^i}(y) \} \} \} \} = \\
 &= \max_{1 \leq i \leq n} \{ \min \{ \min \{ \sup_{x_1} \min \{ \mu_{A_1}(x_1), \mu_{X_1^i}(x_1) \}, \dots \}, \mu_{Y^i}(y) \} \} = \\
 &= \max_{1 \leq i \leq n} \mu_{D \circ R_i}(y)
 \end{aligned}$$

or equivalently :

$$\begin{aligned}
 \mu_{D \circ R_i}(y) &= \min \{ \min_j \tau_{ij}, \mu_{Y^i}(y) \} = \min \{ \tau_i, \mu_{Y^i}(y) \} \\
 \mu_{D \circ R}(y) &= \max_i \mu_{D \circ R_i}(y).
 \end{aligned}$$

Mamdani implication with S=max és T=dot:

$$\begin{aligned}
 \mu_{D \circ R_i}(y) &= \left(\prod_j \tau_{ij} \right) \cdot \mu_{Y^i}(y) = \tau_i \cdot \mu_{Y^i}(y) \\
 \mu_{D \circ R}(y) &= \max_i \mu_{D \circ R_i}(y).
 \end{aligned}$$

where $\max_i \mu_{D \circ R_i}(y)$ can be computed easily based on the known values of $\mu_{D \circ R_i}(y)$.

Summary!

Mamdani type max-min inference algorithm

- 1) Evaluate the value of τ_{ij} , $j = 1, \dots, N$ and τ_i for each rule R_i , and evaluate the value of function $\mu_{D \circ R_i}(y)$ for each y :

$$\tau_{ij} = \begin{cases} \mu_{X_j^i}(x_j^*), & \text{if } x_j \text{ is crisp} \\ \sup_{x_j} \min \left\{ \mu_{A_j}(x_j), \mu_{X_j^i}(x_j) \right\}, & \text{otherwise} \end{cases}$$
$$\tau_i = \min_j \tau_{ij}$$

- 2) Evaluate $\mu_{D \circ R_i}(y)$ for each y :

$$\mu_{D \circ R_i}(y) = \begin{cases} \mu_{Y^i}(y), & \text{if } \mu_{Y^i}(y) \leq \tau_i \\ \tau_i, & \text{otherwise} \end{cases}$$

- 3) Evaluate the membership function of the output signal of the fuzzy system with max operation for all y

$$\mu_{D \circ R}(y) = \max_i \mu_{D \circ R_i}(y)$$

Mamdani tipusú max-dot inference algorithm:

- 1) Evaluate the value of τ_{ij} , $j = 1, \dots, N$ and τ_i for each rule R_i , and evaluate the value of function $\mu_{D \circ R_i}(y)$ for each y :

$$\tau_{ij} = \begin{cases} \mu_{X_j^i}(x_j^*), & \text{if } x_j \text{ is crisp} \\ \sup_{x_j} \left\{ \mu_{A_j}(x_j) \cdot \mu_{X_j^i}(x_j) \right\}, & \text{otherwise} \end{cases} \quad (\text{Changed!})$$

$$\tau_i = \tau_{i1} \wedge \tau_{i2} \wedge \dots \wedge \tau_{iN} = \tau_{i1} \cdot \tau_{i2} \cdot \dots \cdot \tau_{iN} \quad (\text{Changed!})$$

- 2) Evaluate $\mu_{D \circ R_i}(y)$ for each y :

$$\mu_{D \circ R_i}(y) = \tau_i \wedge \mu_{Y^i}(y) = \tau_i \cdot \mu_{Y^i}(y)$$

- 3) Evaluate the membership function of the output signal of the fuzzy system with max operation for all y

$$\mu_{D \circ R}(y) = \max_i \mu_{D \circ R_i}(y)$$

Simplified sum-dot inference algorithm:

- 1) Evaluate the value of τ_{ij} , $j = 1, \dots, N$ and τ_i for each rule R_i , and evaluate the value of function $\mu_{D \circ R_i}(y)$ for each y :

$$\tau_{ij} = \begin{cases} \mu_{X_j^i}(x_j^*), & \text{if } x_j \text{ crisp} \\ \sum_{x_j} \prod \left\{ \mu_{A_j}(x_j), \mu_{X_j^i}(x_j) \right\}, & \text{otherwise} \end{cases}$$

$$\tau_i = \tau_{i1} \wedge \tau_{i2} \wedge \dots \wedge \tau_{iN} = \tau_{i1} \cdot \tau_{i2} \cdot \dots \cdot \tau_{iN}$$

- 2) Evaluate $\mu_{D \circ R_i}(y)$ for each y :

$$\mu_{D \circ R_i}(y) = \tau_i \wedge \mu_{Y^i}(y) = \tau_i \cdot \mu_{Y^i}(y)$$

- 3) Evaluate the membership function of the output signal of the fuzzy system with sum operation for all y

$$\mu_{D \circ R}(y) = \bigvee_i \mu_{D \circ R_i}(y) \approx \sum_i \mu_{D \circ R_i}(y) = \sum_i \tau_i \cdot \mu_{Y^i}(y). \quad \textbf{(Changed!)}$$

if the defuzzified value of the output is computed by the center of gravity (COG) method which is also known as the center of area (COA) method, then

$$y^* = \frac{\int y \mu_{D \circ R}(y) dy}{\int \mu_{D \circ R}(y) dy} = \frac{\int y \sum_i \tau_i \mu_{Y^i}(y) dy}{\int \sum_i \tau_i \mu_{Y^i}(y) dy} = \frac{\sum_i \tau_i y_i^* \int \mu_{Y^i}(y) dy}{\sum_i \tau_i \int \mu_{Y^i}(y) dy} = \frac{\sum_i (\tau_i \int \mu_{Y^i}(y) dy) y_i^*}{\sum_i (\tau_i \int \mu_{Y^i}(y) dy)} = \frac{\sum_i \tau_i^* y_i^*}{\sum_i \tau_i^*}$$